

C1 extends C2

Program Code:

```
package core_java;

class C2 {
    private int a = 10;

    void msg2() {
        System.out.println("Message from C2");
    }

    public int getA() {
        return a;
    }
}

class C1 extends C2 {
    void msg1() {
        System.out.println("Message from C1");
        System.out.println("Value of a: " + getA());
    }
}

public class C1toC2 {
    public static void main(String[] args) {
        C1 obj = new C1();
        obj.msg2();
        obj.msg1();
    }
}
```

Explanation:

This program demonstrates class inheritance. C1 extends C2, so an object of C1 can access the methods of both C1 and C2. The variable a is private in C2 and is accessed through the public getA() method.

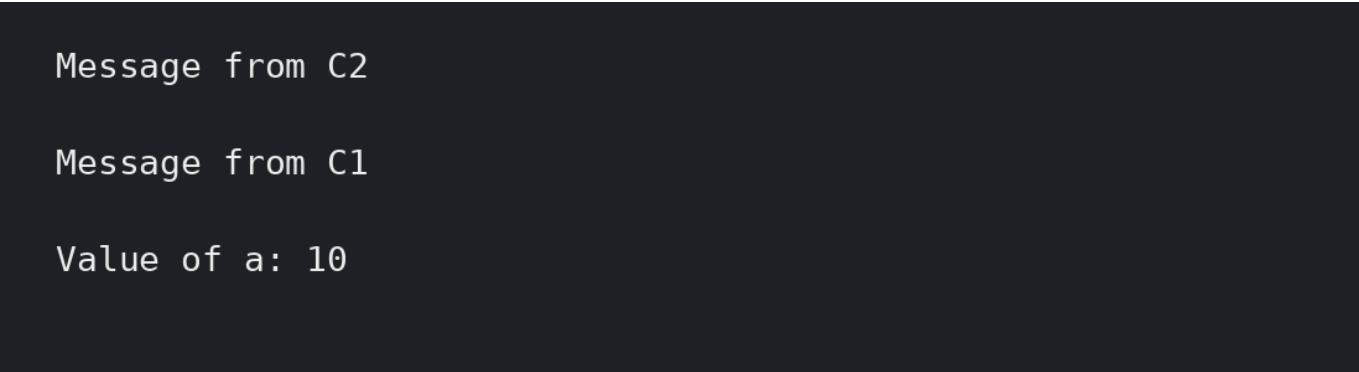
Compilation and Execution Commands:

Save the file as: C1toC2.java

Compile:
javac -d . C1toC2.java

Run:
java core_java.C1toC2

Output Screenshot:



```
Message from C2

Message from C1

Value of a: 10
```

Verification:

The output shows that the object of C1 successfully calls msg2() from C2 and msg1() from C1. The value of the private variable a is correctly obtained through getA().

Result:

Thus, the program was successfully executed and the inheritance relationship C1 extends C2 was verified successfully.